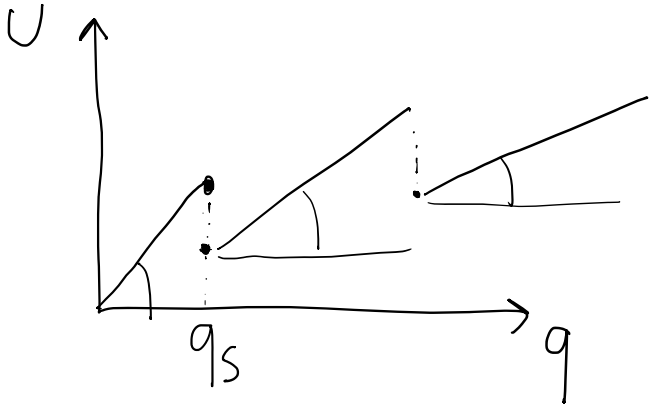
CURVA CARATTERISTICA DEL PRODOTTO



Esempio

Stazione i	M	t_i [min/pezzo]
1	Torno	1.02
2	Rettifica	0.45
3	⋮	0.4
4	⋮	0.923
5	⋮	0.923
6	⋮	1.5
7	⋮	0.75

$m_i = 1$

$\forall i = 1, \dots, 7$

$$q_{1S} = \frac{60 \cdot m_i}{t_i} = 59 \ [\text{pezzi}/\text{ore}]$$

$q_{2S} = 133$; $q_{3S} = 150$; $q_{4S} = 65$; $q_{5S} = 65$;

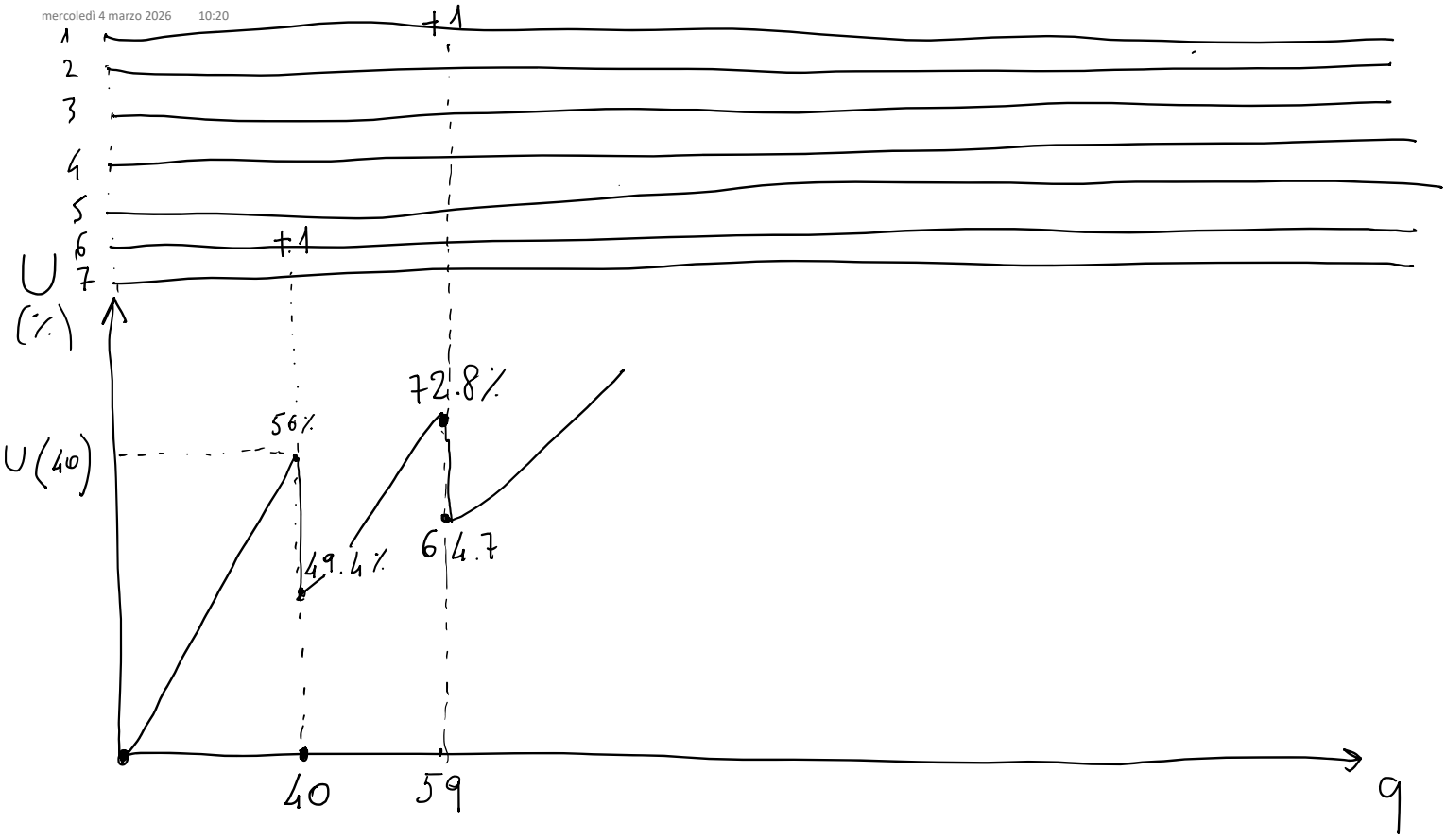
$q_{6S} = 40$; $q_{7S} = 80$

- ordinamento q_S

$q_{6S} \mid q_{1S} \mid q_{4S}, q_{5S} \mid q_{7S}, 2q_6 \mid 2q_{1S} \mid 3q_{6S} \mid 2q_{4S}, 2q_{5S} \mid$

$q_{2S} \mid q_{3S}$

mercoledì 4 marzo 2026 10:20



Esercizio linea-reparti

P1 ▽ → (A) → (B) → (A) → ▽

60 [sec/pezzo] 120 5

P2 ▽ → (A) → (C) → (B) → [D] → (C) → ▽

90 90 60 30 5

P3 ▽ → (B) → (E) → [D] → ▽

60 60 90

L'impianto lavora su 1 turno da 8 ore/giorno

$Q_1 = 800$ [pezzi/giorno] ; $Q_2 = 1200$; $Q_3 = 1000$

$K_1 = 0.98$; $K_3 = 1$; $K_2 = \frac{UT}{UT+DT} = 0.98$

$K_4 = \frac{PT}{UT}$ → 0.83 (reparti) [0.65 ÷ 0.85]

→ 0.94 (linea) [0.80 ÷ 0.95]

Si chiede di dimensionare le linee e i reparti

Layout linea

P1 Stazione A

$$N_A = \frac{800 \times 60}{0.9 \times \underbrace{8 \times 60 \times 60}} \qquad \left(N_{ij} = \frac{Q \times t_{ij}}{\eta \times \dots} \right)$$

$$\eta = K_1 \cdot K_2 \cdot K_3 \cdot K_4 = 0.98 \cdot 0.98 \cdot 0.94 \simeq 0.9$$

$$= 1.85 \Rightarrow 2 \text{ MACCHINE}$$

Stazione B

$$N_B = \frac{800 \times 120}{0.9 \times 8 \times 60 \times 60} \qquad 120 = t_{ij} \qquad = 3.7 \rightarrow 4 \text{ MACCHINE}$$

Stazione A

$$N_A = \frac{800 \times 5}{0.9 \times 8 \times 60 \times 60} = 0.15 \rightarrow 1 \text{ MACCHINA}$$

$$u_A = \frac{1.85}{2} = 0.93 \; ; \quad u_B = \frac{3.7}{4} = 0.93 \; ; \quad u_{A1} = \frac{0.15}{1} = 0.15$$

	Stazione	#	u
P2	A (I)	5	0.83
	C (II)	5	0.83
	B (III)	3	0.93
	D (IV)	2	0.69
	C (V)	1	0.23
P3	B (I)	3	0.77
	E (II)	3	0.77
	D (III)	4	0.87

Layout per reparti → 5 reparti (A, B, C, D, E)

$$M_J' = \sum_{i=1}^{m} \frac{Q_{iJ} \times T_{iJ}^*}{\eta \cdot N_{iJ}} = \text{Tempo a disposizione}$$

$$M_A' = \frac{Q_1 \times T_{A1}^* + Q_2 \times T_{A2}^* + Q_1 \times T_{A1'}^*}{0.8 \times (8 \times 3600)} = 6.94$$

$$\eta = K_1 \cdot K_2 \cdot K_3 \cdot K_4 = 0.98 \times 0.98 \times 0.83 = 0.8$$

$$M_B' = \frac{800 \times 120 + 1200 \cdot 60 + 1000 \cdot 60}{0.8 \times (8 \times 3600)} = 9.90$$

$M_C' = 4.95$; $M_D = 5.47$; $M_E = 2.60$

$u_A = \frac{6.94}{7} = 0.99$; $u_B = 0.99$; $u_C = 0.99$; $u_D = 0.91$

$u_E = 0.87$